Run-length regularity V_2 at the x-mask

Anatomy of v_2, what pairing and products prove, and why the MED-CAP chain freezes here

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Abstract

The length invariant X_n of a folded sign field reduces MED-CAP to a single edge-return. The named remainder V_2 forbids a 6-plateau of x-order run lengths after three consecutive singletons at the 20% x-mask. This note records the anatomy of that colouring, proves the combinatorial half that can be proved, and applies the hydra ward. The factor v_2 is the degree-$(N-2)$ arithmetic-comb chain polynomial, evaluated on the smooth-border nodes; it carries source arithmetic through the Jacobi coefficients of the bordered three-term recurrence. The factor w is the arm sign of the explicit smooth fold, independent of primes. Product combinatorics: a product flips exactly where one factor flips. A 2-regular factor does *not* imply V_2 (the tempting reduction is false). A Chebyshev colouring after the x-mask never ends $(1,1,1)$, and Chebyshev times a flip-first Nyquist burst of w never violates V_2. Random independent products *do* violate, so V_2 is not a theorem of abstract paired colourings. The pattern $(1,1,1)$ produces the dangerous θ-prefix $(1,2)$ if and only if the run count is odd. On the 134-window surface V_2 holds (0 violators); exactly two colouring triples occur, both regular, only one odd (the known saturator χ_4 at index 53, w-driven). The candidate reductions — Jacobi-near-Chebyshev / Freud spacing of v_2, or a further named condition V_3 — are not strictly more elementary than V_2 (they are statements about the same chain polynomial, or about its coefficients, which are the north star of the program). Verdict REGRESS_DETECTED: the escape $T_2' \Leftarrow \cdots \Leftarrow V_2$ freezes at V_2. The pointwise floor $g_i \geq 3/8$ is a theorem of the construction *modulo* V_2. T1 is unchanged. No RH claim.

Status. Lemmas 1–4 are proved (product XOR, pairing parity). Lemma 2 is a machine-checked negative (the 2-regular reduction is false). Lemma 5 is a finite machine census of trigonometric colourings, not a cofinal law. Proposition 1 is a finite census of the construction, not a cofinal law. The hydra ward of §6 fires: REGRESS_DETECTED. V_2 is the final named lemma of the MED-CAP chain. No further stage is opened. T1 is not touched. No RH claim.

1 Recollection

We use the sign-field notation of [2] without change. Blocks of the source are paired consecutive same-sign runs of $c_t = w\,x\,v_2\,\phi$ on the bulk θ-grid after a 20% x-hull mask, pairing from x-min, odd tail its own block. Condition V_2 of [2] §5: if the last three x-order run lengths R_{mask} of $\operatorname{sign}(v_2\,w\,x)$ are $(1,1,1)$, then among the four preceding run lengths at least two are ≤ 2. Equivalently, no 6-plateau at the mask after a triple singleton. The minimal grid-consistent violator is the x-order

$$(\underbrace{3,\ldots,3}_{8},\ 1,\ 1,\ 1),$$

which pairs to θ-order $n = (1,2,6^4)$ and violates MED-CAP at the edge (short C_2). A spike $(1,2,6,4,\ldots)$ saturates and does not violate.

The chain after [1, 2] is

$$T_2' \Longleftarrow g_i \geq \tfrac{3}{8} \Longleftarrow \text{MED-CAP} \Longleftarrow X_n \Longleftarrow V_2,$$

with X_{tile}, the SEP-Satz, and the pairing lemmas of [2] already proved. This note attacks V_2.

2 Anatomy

2.1 What v_2 is

A window of depth N carries an arithmetic comb (prime-power, Dirichlet, or second-family) and a smooth border comb. The bordered chain of [2] §2 is the three-term recurrence of the *arithmetic* signed measure $\mu - \nu$, with Jacobi coefficients $(\hat{\alpha}_k, \gamma_{k+1})$ computed from that comb; the smooth border enters the pairing scalars ρ_k but not those Jacobi coefficients. The polynomial v_2 is the scaled monic chain value of degree $N-2$, evaluated on the smooth-border nodes. It is therefore an orthogonal polynomial of a discrete signed source measure, sampled on a different (border) subset of the same fold-grid. It is *not* a deterministic function of the mesh/fold geometry alone: the primes enter through $(\hat{\alpha}_k, \gamma_{k+1})$. Terminal coefficients on the 134-window surface sit near Chebyshev ($\max_k |\gamma_k - \frac{1}{4}| \leq 0.038$, median 0.012 among the last three), but that proximity is measured, not a theorem, and the coefficients *are* the chain — the north-star object of the program.

2.2 What w is

The weight w on a border node is the folded smooth-comb mass of that fold-lag. Its sign is the arm: $\operatorname{sign}(w) = +1$ on μ cells of d_{arm} and -1 on ν cells. The smooth comb is an explicit positive log-grid times the archimedean tent; d_{arm} can change sign by arch-versus-atom cancellation. No primes enter w. On the four $w = 9$ worlds, w has a single sign-run on the whole bulk, so the colouring equals $\operatorname{sign}(v_2)$ times $\operatorname{sign}(x)$ (one flip at $x = 0$, interior to the bulk). At depth, w may oscillate: χ_4 index 53 has fourteen w-runs, with a Nyquist tail at the $+x$ mask.

2.3 What the x-mask is

The bulk keeps atoms with x inside the central 60% of the hull (edge fraction 0.20). x-order is increasing x; the last runs of R_{mask} sit at the $+x$ mask edge ($x \approx +0.60$, $\theta = \arccos x$). Because $x = \cos\theta$ is decreasing in θ on $[0, \pi]$, that edge is θ-min of the bulk, and an odd tail there becomes block 0 in θ-order.

3 What pairing and products prove

Lemma 1 (flip XOR)**.** *Let $A, B \in \{\pm 1\}^n$. The product AB changes sign at index i if and only if exactly one of A, B changes sign at i. In particular, a simultaneous flip of both factors does* not *end a product-run (in-phase locking).*

Proof. $(AB)_i = A_i B_i$, so the sign ratio $(AB)_i/(AB)_{i-1}$ equals -1 precisely when exactly one factor ratio is -1. □

Lemma 2 (2-regular does not imply V_2)**.** *There exist colourings in which one factor is 2-regular (all runs of length 1 or 2) and the product nevertheless violates V_2. Among 8000 random such products, a positive fraction violate (companion script, gate G2).*

The tempting reduction "v_2 Chebyshev-class $\Rightarrow V_2$" is therefore not a theorem of product combinatorics. It is also false as a hypothesis on the source: v_2 has bulk-mode 2 but admits runs of length 6 at the mask (already on the $w = 9$ FRAME-A window).

Lemma 3 (Chebyshev mask)**.** *A colouring* $\operatorname{sign}(\sin m\theta)$ *on a uniform θ-grid, after the same* 20% *x-mask, never ends with three consecutive singleton runs. (Re-gated from [2] Lemma 5.)*

Lemma 4 (parity of $(1, 1, 1)$)**.** *Let $R = (r_0, \ldots, r_{\nu-1})$ end with $(1, 1, 1)$. Pairing from x-min, the θ-order prefix is $(n_0, n_1) = (1, 2)$ if and only if ν is odd. If ν is even, then $n_0 = 2$.*

Proof. The odd tail exists only for odd ν, and equals $r_{\nu-1} = 1$; the last complete pair is then $(r_{\nu-3}, r_{\nu-2}) = (1, 1)$, summing to 2. Reverse to θ-order. If ν is even there is no tail: the last pair is $(1, 1)$, which becomes $n_0 = 2$. □

Thus a colouring triple is load-bearing for X_n only when the run count is odd. An even triple cannot create the unique small-separation locus of [2] Lemma 3.

Lemma 5 (Chebyshev times a Nyquist burst)**.** *On* 425 *trigonometric windows ($T \in \{80, \ldots, 400\}$, five degrees, burst lengths* 3 *through* 7*), the product of a Chebyshev colouring after the x-mask with a flip-first Nyquist burst of w at the $+x$ edge produces* 0 *tails* $(1, 1, 1)$ *and* 0 *V_2 violations. A constant w times a v_2 that itself ends $(2^8, r, r, r, 1, 1, 1)$ with $r \geq 3$ satisfies V_2 (the four preceding include two bulk* 2*s) and pairs to a spike, not a plateau.*

The burst is the natural adversary for a w-driven triple: monochrome bulk, then three alternating cells. Against a 2-regular factor it does not produce the dangerous pattern. Random independent run-products, by contrast, violate V_2 in a few per thousand draws: the obstruction is not abstract product combinatorics.

4 The named violator, precisely

The x-order $(3^8, 1, 1, 1)$ has odd length, pairs to $n = (1, 2, 6^4)$, fails V_2 (prev4 $= (3, 3, 3, 3)$), and fails MED-CAP. The shorter ending $(2^8, 3, 3, 1, 1, 1)$ satisfies V_2 (prev4 $= (2, 2, 3, 3)$) and pairs to $n = (1, 2, 6, 4, \ldots)$, whose third-of-five is 4 (saturating spike). Degree counting does not forbid the plateau: a degree-$(N - 2)$ polynomial may cluster zeros. Fold multiplicity ≤ 2 and the mesh identity constrain occupation, not v_2-run lengths.

5 The source surface

Proposition 1 (one-shot census)**.** *On the* 134 *windows of [1] Proposition 5 (*42 *prime-power core* $+ 8$ *second-family* $+ 42 + 42$ *Dirichlet; the same surface as [2] Proposition 8):*

1. *V_2 holds on every window (*0 *colouring violators). Chebyshev colourings on the same hulls:* 0 *violators. Chebyshev times the actual w:* 0 *violators.*

2. *Colouring triples* $(1, 1, 1)$ *occur twice, both regular:*

   - *χ_3 index* 16*: v_2 itself ends* $(1, 1, 1)$*, w is constant on the mask neighbourhood (three w-runs globally, a length-*1 *stub far from the $+x$ edge), prev4* $= (3, 2, 1, 3)$*, even run count, θ-prefix starting $n_0 = 2$ (Lemma 4: not small-sep).*
   - *χ_4 index* 53*: the unique* $(1, 2)$ *of [2]. The triple is w-driven (v_2 last three* $=$ $(3, 1, 4)$*, w last three* $= (1, 1, 1)$*), prev4* $= (4, 1, 2, 4)$*, odd run count, saturating prefix* $(1, 2, 6, 5, 4, 4)$*.*

3. *v_2-only triples: one (the χ_3 index 16 case). w-only triples: seven. Six of those seven do not produce a colouring triple (the Chebyshev-times-burst cancellation of Lemma 5, realised by the source).*

4. *On the four $w = 9$ worlds, w has one bulk sign-run and there is no colouring triple. Weights on the mask tail satisfy $|w| \geq 1.2 \cdot 10^{-5}$ (not a floating-sign artefact).*

5. *Prefix $(1, 2)$ occurs once, as in [2].*

So V_2 holds on every window the construction currently produces, including the unique load-bearing triple. A deeper window is not excluded by this census.

The χ_3 index-16 window is the existence proof that v_2 *can* triple at the mask. One cannot prove "v_2 never ends $(1, 1, 1)$". What remains is exactly V_2: when a triple occurs, the four preceding runs are mixed.

6 Hydra ward

The charter: if a proof of V_2 only reduces it to a further condition, that condition must be strictly more elementary (closer to the explicit construction) than V_2. Otherwise the verdict is REGRESS_DETECTED and the escape freezes at V_2, with the complete conditional $T_2' \Leftarrow \cdots \Leftarrow V_2$. No further stage is opened.

Candidate reductions, and why they fail the elementary test:

1. *v_2 is a mesh/fold function.* False: v_2 is the arithmetic OP (Section 2). Case (i) of the charter does not fire.

2. *A named p^k-constellation is mesh-excluded.* The violator of §4 is legal as a $\pm$ colouring, legal as a pairing, and legal as the restriction of a degree-$(N-2)$ polynomial. No finite construction identity plus named classical theorem forbids it. It is not realised on the 134-window surface; that is a census, not an exclusion.

3. *Freud / arcsine spacing, or Jacobi $\gamma_k \to 1/4$.* v_2 is not the OP of a fixed positive continuous measure. The Jacobi coefficients of the discrete signed comb *are* the bordered chain. Reducing V_2 to a Chebyshev-asymptotic for those coefficients is a reduction toward the north star (L*, the wall), not toward the fold/mesh. Not more elementary.

4. *Replace V_2 by "v_2 is 2-regular at the mask".* Lemma 2 refutes the combinatorial implication even as an abstract product statement, and the replacement is false on the surface (runs of length 6). Not more elementary, and not true.

5. *Replace V_2 by a condition on w alone.* The load-bearing triple at χ_4 index 53 is w-driven, and w is construction-explicit. But the MAIN family (single-arm w) reduces V_2 back to v_2-only, and χ_3 index 16 is a v_2-triple with w constant at the mask. A w-only lemma does not cover the family that carries the floor.

None of these is a strictly more elementary lemma on which V_2 would rest. Opening a V_3 would be the hydra. The ward fires.

7 The frozen chain

Proposition 2 (conditional floor, frozen)**.** *On every tiled packing produced by the construction of [2] §2,*

$$X_{\text{tile}} + \textit{pairing lemmas of [2]} + V_2 \quad\Longrightarrow\quad X_n \quad\Longrightarrow\quad \text{MED-CAP} \quad\Longrightarrow\quad g_i \geq \tfrac{3}{8} \quad\Longrightarrow$$

V_2 is the final named lemma of this chain. It is verified on the 134*-window surface and is not proved as a cofinal law of the bordered chain. The cubic-mass bound of [1] §T1 carries, still carrying C_K.*

Unconditional on the present surface: X_n, MED-CAP, the pointwise floor, and T_2'. Cofinal: the same statements modulo V_2.

8 T1

Unchanged. V_2 is a constraint on run lengths of a sign field. The masses q_i see the same field as an amplitude. Proving V_2, even cofinally, does not produce a family-uniform C_K. The r358 record (per-family freeze maximum fails on the test set; ceiling 23.70 against freeze 4.91) stands.

9 Machine checks

Every numbered lemma of §3 and the violator identities of §4 are re-proved by exact arithmetic in the companion script `verify_v2_steps.py` (same directory; twelve gates G1–G12), together with the three construction pins of Proposition 1 (ii)–(iv) on six windows (four $w = 9$, χ_4 index 53, χ_3 index 16). The 134-window census is a one-shot enumeration against the same builders (not a sealed probe), recorded here. Exit line: `V2 STEPS VERIFIED`.

Claim boundary

Finite identities on explicit colourings, a pairing-parity lemma, a product-containment lemma, and a named freeze of an analytic remainder V_2 on a bordered-chain polynomial. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

[1] S. Hamann, *A median cap for gaps of a tiled integer packing*, standalone note, August 2026, `rh/problem/medcap_lemma.tex`.

[2] S. Hamann, *The length invariant X_n of a folded sign field*, standalone note, August 2026, `rh/problem/xn_invariant.tex`.